3.1.2 Group 2

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Redox reactions and reactivity of Group 2 metals		
(a)	the outer shell s^2 electron configuration and the loss of these electrons in redox reactions to form 2+ ions	
(b)	the relative reactivities of the Group 2 elements Mg → Ba shown by their redox reactions with: **(i)** oxygen (ii) water (iii) dilute acids	Reactions with acids will be limited to those producing a salt and hydrogen.
(c)	the trend in reactivity in terms of the first and second ionisation energies of Group 2 elements down the group **(see also 3.1.1 c)**	*M3.1* Definition of second ionisation energy is **not** required, but learners should be able to write an equation for the change involved.
Reactions of Group 2 compounds		
(d)	the action of water on Group 2 oxides and the approximate pH of any resulting solutions, including the trend of increasing alkalinity	
(e)	uses of some Group 2 compounds as bases, including equations, for example (but not limited to): **(i)** $Ca(OH)_2$ in agriculture to neutralise acid soils (ii) $Mg(OH)_2$ and $CaCO_3$ as 'antacids' in treating indigestion.	